

Saket Upperla

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EDUCATION

University of Texas at Austin

Aug. 2023 - May 2027

B.S. Computer Science | **GPA: 3.85/4.0**

- **Courses:** Data Structures, Operating Systems, Computer Architecture, Algorithms & Complexity, Principles of Machine Learning, Discrete Math, Multivariable Calculus, Linear Algebra, Probability & Statistics

EXPERIENCE

Software Engineer Intern, Jump Finance

July. 2025 - Present

- Built functionality enabling updates to **100+** active student business-loan records from West African students via a **RESTful API**, ensuring secure backend storage with **Python (FastAPI)**, **MySQL**, and **DBeaver**.
- Designed and launched a student referral program with targeted incentives, boosting membership by **12%**.

Software Engineer Intern, Livegate

Jan. 2025 - May 2025

- Led biweekly sprints to design and deploy a scalable sports communication backend with **Node.js/Express** on **Google Cloud Platform** with **10+ REST endpoints** for authentication, profiles, and messaging.
- Designed **MongoDB** profile/game database, containerized a **Selenium** NBA/MLB/NFL/NCAA scraper with **Docker** (scheduled via **Cloud Scheduler**), and implemented a **Socket.io** chat layer for live communication.

Undergraduate Researcher, UT Robot Intelligence Lab

Sept. 2024 - Jan. 2025

- Led an undergraduate team to enhance the **GraphIRL** framework in **Python**, fixing **dependency issues** and **memory leaks**, and leveraging **CUDA GPUs** for intensive graph generation to prepare it for open-source release
- Enhanced the **GraphIRL** reward function with depth-perception features using **NumPy** and **PyTorch**, and evaluated in **RoboSuite** on Reach, Push, and Peg-in-Box tasks, boosting success rate by **19.2%**.

PROJECTS

CampAIgn | *Type Script, Canva App SDK, REST APIs*

- Created an AI-powered Canva app that auto-generates tailored campaign messages for email, Twitter, and LinkedIn, helping marketers publish faster without leaving the Canva platform, reducing publishing time by **38%**.
- Integrated **OAuth 2.0** and **REST APIs** for LinkedIn, Mailchimp & Twitter to enable one-click publishing.

Storytunes | *Next.js, Express.js, Spotify API, Axios*

- Built an app enabling users to discover theme-based songs to pair with photos on their Instagram Stories, featuring a **Next.js** frontend, **Express.js** backend, and integration with the **Spotify API**.
- Generated **200+** song suggestions per theme, reducing song-selection time by **17%** based on user feedback.

TempoBuddy | *React, TypeScript, MediaPipe Pose (WASM), FFmpeg, librosa, SVG*

- Launched a browser dance timing coach that compares user routines to a reference and quantifies on-beat accuracy; improved on-beat accuracy by **24%** and earned dance-team acceptance after **2** prior rejections.

PintOS | *C, x86 Assembly, Threads, Virtual Memory, File Systems*

- Built an OS kernel with **20+** system calls, argument parsing, ELF binary stack setup, and a custom **file system**.
- Implemented **virtual memory**, demand paging with supplemental page tables, frame/swap management, dynamic **stack growth**, **indexed file allocation**, subdirectory support, and **concurrency** control with locking.

Adoption Shelter Classifier | *Python, NumPy, pandas, TensorFlow, scikit-learn, XGBoost, LightGBM, Optuna*

- Built and tuned an adoption outcome classifier on **100,000+** dog/cat shelter records, using **scikit-learn**, **XGBoost**, **LightGBM**, and **Optuna** to achieve a balanced accuracy of **0.485** on **27,000+** holdout cases.

AWARDS

59th globally (20th in the U.S.) out of 12,000+ teams @ IMC Prosperity 3 | *Python, Algorithmic Trading*

- Engineered trading algorithms by utilizing market-making, mean-reversion, moving-average signals, pairs-trading, Black-Scholes option-pricing, and bot-trade tracking techniques to maximize simulated PnL.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL, Python, C, C++, TypeScript, HTML/CSS

Tools: Docker, Kubernetes, Node.js, React, TensorFlow, PyTorch, scikit-learn, Git, Postgres, MongoDB, AWS, GCP

Concepts: Object-Oriented Design, Distributed Systems, Cloud Computing, CI/CD, Agile Development